

Bayesian Inference and Tossing Coins

Narnia Poddar

1 What is Bayesian Inference?

We have all spent countless hours, or so it feels, working on a problem regarding a fair coin toss, fair die roll, or a fair draw of cards. But how do we know if the coin being tossed is a fair coin or not? Bayesian inference is a technique that uses Bayes' theorem to help us work towards answering questions of this nature.

2 Bayes' Theorem

Bayes' theorem states that the conditional probability of A given B (the probability of event A given the occurrence of event B) can be calculated from $P(B|A)$, $P(A)$, and $P(B)$ as follows:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

2.1 Proof of Bayes' Theorem

To prove Bayes' theorem, we begin with the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Multiplying both sides by $P(B)$ gives us:

$$P(B)P(A|B) = P(A \cap B)$$

Similarly, we can use the definition of the conditional probability for $P(B|A)$:

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

And multiplying both sides by $P(A)$:

$$P(A)P(B|A) = P(B \cap A).$$

However, since $P(A \cap B) = P(B \cap A)$, we can equate the right-hand sides of the equations:

$$P(B)P(A|B) = P(A)P(B|A)$$

Dividing both sides by $P(B)$ gives us Bayes' theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

3 Coin Tossing Example

Suppose we want to determine the fairness of a coin – we want to know the probability of getting heads after tossing the coin. Assume that the coin always lands on either heads or tails, and we have no existing observations about the outcomes of flipping the coin. There are multiple ways to approach this problem: either using the frequentist approach (estimates of probability are gathered after repeating the experiment numerous times) or using Bayesian inference (start with a belief about the probability and keep updating it using Bayes’ Rule).

3.1 Determining the Probability Using the Frequentist Approach

In the frequentist view, the “probability” of an event refers to the proportion of times the event occurs if infinitely many trials are executed:

$$\lim_{n \rightarrow \infty} \frac{n_A}{n},$$

where n_A is the number of occurrences of event A in n trials. In reality, a statistician cannot run infinitely many trials. What a statistician can obtain is an estimate for $P(A)$: $P(A) = \frac{n_A}{n}$, which will get closer to $P(A)$ as n gets larger. In other words, the error (the difference between $P(\hat{A})$ and the true value) associated with our estimate of $P(\hat{A})$ becomes smaller as n becomes large.

The frequentist interpretation of the coin-flipping problem doesn’t make any assumptions about the probability of observing a head or tail before actually flipping the coin – we can’t use our existing knowledge to incorporate a belief that most coins are usually fair or close to fair. To obtain an accurate estimate of the probability of the coin landing on heads, this approach requires tossing the coin a fairly large number of times.

We run into pitfalls with this approach if we can only flip the coin a few times. Although we can normally flip a coin as many times as desired, other statistical problems may have limits on how many trials can be conducted. For example, say we want to determine the probability of a candidate winning an election. The event may only occur a few times, and the observer has no control over how many trials occur. So what can we do when the frequentist approach fails?

3.2 How is Bayesian Inference Different?

In Bayesian inference, the “probability” of an event is considered to be the subjective belief an observer holds about the likelihood that an event would occur. In this point of view, it is fine to have a view about the probability of an event for which no trials have even been conducted. It also allows for multiple observers to hold different beliefs about the probability distribution of the same event.

In this framework, one starts with a subjective belief about the probability of an event, called the prior. As one observes the outcomes of trials, they update their beliefs to incorporate the information obtained from the trials. The updated belief is known as the posterior. The update is done using Bayes Theorem and the process is called Bayesian inference (or Bayesian Learning).

3.3 How is Bayes’ Theorem Used to Update Beliefs?

It is customary in Bayesian Inference to write Bayes’ theorem in the following form:

$$P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)}$$

Here,

- θ represents the object that we are interested in learning about (which in our case would be the probability of the coin landing on heads when it is tossed).
- X represents the observations that we have collected (which in our case would be the sequence of heads and tails observed after the coin was tossed a certain number of times).
- $P(\theta)$, called the prior, is our belief about the probability of the event (landing on heads) before any coin tosses are made.
- $P(X|\theta)$, called the likelihood, is the probability of observing the outcome X given a specific θ – for example, given that the probability of the coin landing on heads is 0.4.
- $P(\theta|X)$, called the posterior, is our updated belief about probability of θ , given the sequence of heads and tails (X) that we have observed so far.

$P(X)$, the unconditional probability of observing X based on prior beliefs, can be computed as a summation of the conditional probabilities $P(X|\theta)$ over all possible distinct values of $\theta \in 1, 2, \dots, n$ (in other words all possibilities based on our prior beliefs),

$$P(X) = \sum_{i=1}^n P(X|\theta_i)P(\theta_i).$$

So we start with a prior belief about the probability distribution of the coin landing on heads when tossed. Then, applying Bayes' Theorem, we update our belief about the probability distribution based on the actual observations collected. Note that when the number of observations becomes sufficiently large, the posterior probability estimate obtained by Bayesian inference would normally converge to a frequentist estimate of the probability.

4 Applying Bayesian Learning in a Discrete Setting

In this section, we explore the application of Bayesian learning to the coin tossing problem where the probability of landing on heads can only take a discrete set of values. For example, say that the probability of landing on heads is either 0.4 or 0.6. So, the coin is biased, but we do not know which has a higher probability: heads or tails. In our prior, we assume both cases to be equally likely:

$$P(\theta = 0.4) = 0.5$$

$$P(\theta = 0.6) = 0.5$$

where θ represents the probability that the coin lands on heads when tossed. Suppose we toss the coin N times and observe h heads. This is our set of observations, X . Then, given a probability θ of observing heads in a single coin toss, and assuming outcome of each toss is independent of other tosses, we can compute the likelihood of h heads in N tosses as:

$$P(X|\theta) = \binom{N}{h} \theta^h (1 - \theta)^{N-h}$$

and

$$P(X) = \sum_{\theta} \binom{N}{h} \theta^h (1 - \theta)^{N-h} \cdot P(\theta)$$

so, our Bayesian posterior belief is given by:

$$P(\theta|X) = \frac{\binom{N}{h} \theta^h (1 - \theta)^{N-h} \cdot P(\theta)}{\sum_{\theta} \binom{N}{h} \theta^h (1 - \theta)^{N-h} \cdot P(\theta)}$$

which lends itself to the intuitive interpretation that the posterior probability of θ is the likelihood of observing h heads in N trials assuming θ is true, multiplied by the prior probability of θ , and divided by the total probability of observing h heads in N tosses given prior beliefs about all possible values of θ .

Specifically let's assume that we toss the coin 5 times and observe 3 heads. We can compute the posterior probabilities $P(\theta = 0.4|h = 3, N = 5)$ and $P(\theta = 0.6|h = 3, N = 5)$ as below:

$$P(h = 3, N = 5|\theta = 0.4) = \binom{5}{3} 0.4^3 (1 - 0.4)^{5-3} = 0.2304$$

$$P(h = 3, N = 5|\theta = 0.6) = \binom{5}{3} 0.6^3 (1 - 0.6)^{5-3} = 0.3456$$

Then, $P(X)$ is:

$$\begin{aligned} P(h = 3, N = 5) &= P(h = 3, N = 5|\theta = 0.4) \cdot P(\theta = 0.4) + P(h = 3, N = 5|\theta = 0.6) \cdot P(\theta = 0.6) \\ &= 0.2304 \times 0.5 + 0.3456 \times 0.5 = 0.2880 \end{aligned}$$

and the posterior beliefs are:

$$\begin{aligned} P(\theta = 0.4|h = 3, N = 5) &= \frac{P(h = 3, N = 5|\theta = 0.4) \cdot P(\theta = 0.4)}{P(h = 3, N = 5)} = \frac{0.2304 \times 0.5}{0.2880} = 0.4 \\ P(\theta = 0.6|h = 3, N = 5) &= \frac{P(h = 3, N = 5|\theta = 0.6) \cdot P(\theta = 0.6)}{P(h = 3, N = 5)} = \frac{0.3456 \times 0.5}{0.2880} = 0.6 \end{aligned}$$

Given that our observations have a higher number of heads compared to tails, our posterior belief now assigns higher odds to the probability of heads being 0.6 instead of the prior of 0.4 as one would expect.

5 Applying Bayesian Learning in Continuous Settings

In a more general setting, we would like to be able to take on any value within a continuous range. For example, in our coin tossing example, we likely want to allow to take on any value in the range $[0,1]$. In this case will be a continuous variable. Probability distributions of continuous random variables are generally characterized using a probability density function (pdf). The density function at any point describes the probability per unit length around that point. Since any continuous range has infinitely many values, the probability of the variable exactly taking on any single value (a point on the range) is zero. Probability is defined on a range $[a, b]$ as:

$$P(a \leq \theta \leq b) = \int_a^b p(\theta) d\theta$$

where p is the density function and P is the probability. Given that probabilities are fully specified by the density function and can be obtained using the above formula, one can work directly with prior and posterior density functions when applying Bayesian inference on problems with continuous variables.

5.1 Bayes' Theorem for Density Functions

Bayes' theorem for pdfs can be written as:

$$p(\theta|X) = \frac{p(X|\theta) \cdot p(\theta)}{p(X)}$$

where $p(\theta)$ denotes the prior density of θ , $p(X|\theta)$ is the conditional density of X given θ , $p(X)$ is the unconditional density of X , and $p(\theta|X)$ is the posterior density of θ given X . Often the relationship between the prior and posterior is represented as a proportionality:

$$p(\theta|X) \propto p(X|\theta) \cdot p(\theta)$$

The posterior density is proportional to the prior density and the likelihood of the observations under the prior density.

When working with Bayesian inference on a continuous θ , it is often beneficial to select pdfs for priors such that the functional form for the posterior has the same functional form as the prior itself. Such a combination of likelihood and priors are called conjugate priors. It allows us to reuse the Bayesian update algebra over and over as we observe new data and continue to update our beliefs. For example, if we observe a new set of trials Y , after having updated our beliefs to $p(\theta|X)$ from $p(\theta)$ using the observations in X , as long as $p(\theta|X)$ has the same functional form as $p(\theta)$, we can make use of the same algebra or code to obtain $p(\theta|X, Y)$ using the observations in Y and using $p(\theta|X)$ as our current prior.

5.2 Bayesian Inference for Coin Tossing with Continuous θ

In this setting we allow for θ to take any value in the range $[0, 1]$. Note that our likelihood function $p(X|\theta)$ is unchanged from earlier, since for discrete variable X :

$$p(X|\theta) = P(X|\theta) = \binom{N}{h} \theta^h (1 - \theta)^{N-h}.$$

A conjugate prior for this likelihood function is the Beta distribution. The density function of the Beta distribution is given by:

$$p(\theta) = \frac{\theta^{a-1} (1 - \theta)^{b-1}}{B(a, b)}$$

where $B(a, b)$ is the Beta function which ensures that the probability that lies in $[0, 1]$ is 1 and $a, b > 0$ are parameters that determine the shape of the distribution (such as where the peak is, how heavy the tails are, etc). To see why the Beta distribution forms a conjugate prior with our likelihood in this case, we observe:

$$p(\theta|X) \propto p(X|\theta) \cdot p(\theta) \propto \theta^{a-1} (1 - \theta)^{b-1}$$

Simplifying, we obtain:

$$p(\theta|X) \propto \theta^{a+h-1} (1 - \theta)^{b+N-h-1}.$$

So, $p(\theta|X)$ has a density function given by a Beta Distribution with parameters $a' = a + h - 1$, and $b' = b + N - h - 1$.

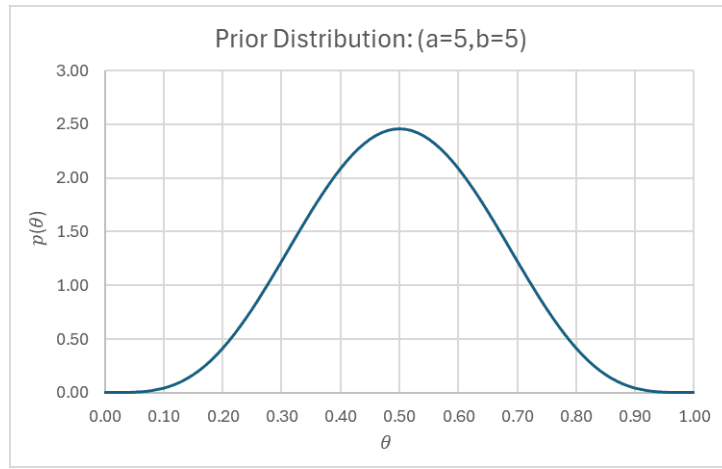
We start with a prior of a Beta distribution with $a = 5, b = 5$.

$$p = \frac{\theta^{5-1}(1-\theta)^{5-1}}{B(5,5)}.$$

This leads to a distribution which peaks at 0.5 but is still quite spread out as shown in the figure below. Under this prior,

$$P(\theta \leq 0.5) = 0.5, P(\theta \geq 0.5) = 0.5.$$

It reflects a belief that while there is substantial uncertainty about the value of θ , we do not have any view on if the coin is fair or unfair.



Now, we update the belief using the observation $X(N = 5, h = 3)$. The posterior will be a Beta distribution with $a' = a + h - 1 = 7$, and $b' = b + N - h - 1 = 6$. In other words

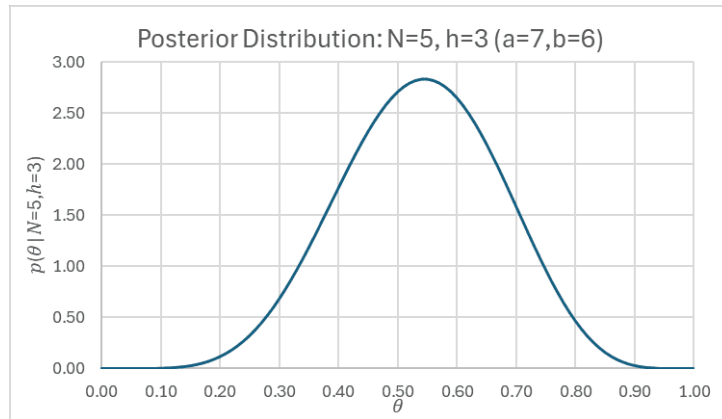
$$p(\theta|N = 5, h = 3) = \frac{\theta^{7-1}(1-\theta)^{6-1}}{B(7,6)}.$$

The plot of the density shows that the peak of the distribution has shifted rightwards from its 0.5 position in the prior and the spread of the distribution has reduced. Under the posterior, the odds of the probability of heads being greater than 0.5 is now higher than odds being lower than 0.5:

$$P(\theta \leq 0.5|N = 5, h = 3) = 0.39$$

$$P(\theta \geq 0.5|N = 5, h = 3) = 0.61$$

The plot below shows the density of the posterior.

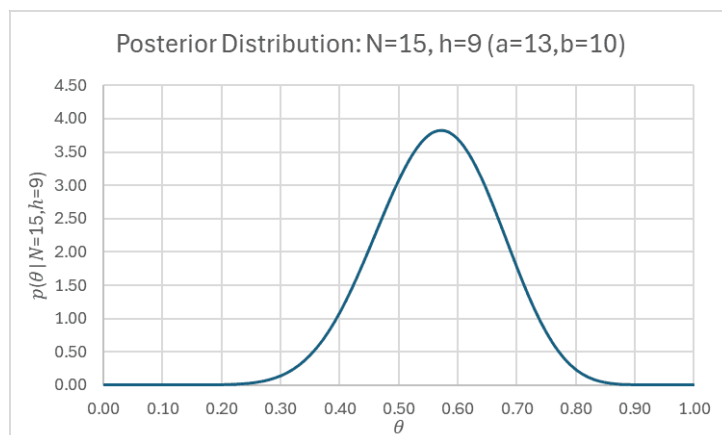


To see how the distribution changes with sample size, we look at another sample with the same proportion of heads but larger sample size: $N = 15, h = 9$. The posterior $p(\theta|N = 15, h = 9)$ is now a Beta distribution with $a' = a + h - 1 = 13$, and $b' = b + N - h - 1 = 10$. The distribution has become narrower and the odds of the probability of heads being greater than 0.5 is now even higher than the $N = 5$ case:

$$P(\theta \leq 0.5|N = 15, h = 9) = 0.26$$

$$P(\theta \geq 0.5|N = 15, h = 9) = 0.74$$

This shows that as the sample size increases, the uncertainty about the value of θ reduces and our belief about starts to center towards the sample estimate.



6 Other Applications

Even though determining the fairness of a coin is something that most people probably don't find themselves needing to do in their daily lives, techniques used for Bayesian inference are useful for a variety of problems. For example, Bayesian inference can be used for updating predictions for weather events based on incoming data or modeling the spread of diseases.

References

Bayesian Inference of a Binomial Proportion - The Analytical Approach. (n.d.). QuantStart.
<https://www.quantstart.com/articles/Bayesian-Inference-of-a-Binomial-Proportion-The>

-Analytical-Approach/.

“Bayesian Statistics: A Beginner’s Guide.” QuantStart,
<https://www.quantstart.com/articles/Bayesian-Statistics-A-Beginners-Guide/>.

Lecture 4: Estimating Probabilities from data. (n.d.). CS@Cornell.
<https://www.cs.cornell.edu/courses/cs4780/2018sp/lectures/lecturenote04.html>.

Lancaster, T. (2004). Introduction to Modern Bayesian Econometrics. Wiley.